

SLIDE 1: 1: Title & Hook

VISUAL SLIDE LAYOUT DESIGN

- Dark mode layout with a sleek corporate theme.
- A split-screen visual showing a live Bengaluru junction on the left and its corresponding Spatial Scene Graph representation on the right.
- Subtitle: Hierarchical Scene Understanding Pipeline (HSUP) for Mixed Traffic Environments.

KEY TECHNICAL & BUSINESS TAKEAWAYS

- Introduce the team and the project.
- Establish the core message: Flat detection is outdated; relational reasoning is the future of automated traffic policing.

SPEAKER SCRIPT & PRESENTATION NOTES

"Good morning, esteemed judges. We are representing the TrafficSentinel team, and today we are excited to present TrafficSentinel AI—a next-generation automated traffic enforcement system built specifically for the chaos, density, and unique challenges of Indian roads. Rather than just identifying boxes, our system reconstructs a logical scene graph of the road, giving traffic authorities like the Bengaluru Traffic Police an unmatched combination of high recall and near-zero false alarms."

SLIDE 2: 2: The Mixed Traffic Challenge

VISUAL SLIDE LAYOUT DESIGN

- 3 columns showing:
 - 1. Density & Heterogeneity: Auto-rickshaws, bikes, cars, and pedestrians sharing tight spaces.
 - 2. Adverse Conditions: Heavy monsoons, water reflection, night low-light, and headlights glare.
 - 3. False-Positive Catastrophe: Flat detectors flagging a pedestrian walking next to a motorcycle as a "rider without a helmet"

KEY TECHNICAL & BUSINESS TAKEAWAYS

- Define the limitations of standard YOLO baselines.
- Highlight the lack of spatial context, lack of temporal awareness, and degradation under adverse weather.

SPEAKER SCRIPT & PRESENTATION NOTES

"If you deploy a standard, off-the-shelf object detector at a Bengaluru junction, it will quickly fail. Why? Because generic models lack context. When a pedestrian walks close to a motorcycle, a flat classifier associates that person with the two-wheeler and generates a false 'no-helmet' challan. In heavy monsoon rains or at night, image quality drops, and OCR systems fail to read plates, resulting in endless manual review overhead. We set out to solve this fundamental contextual gap."

SLIDE 3: 3: The Innovation - HSUP Architecture

VISUAL SLIDE LAYOUT DESIGN

- A clean, horizontal block diagram showing the 5 layers:
- Layer 1: Scene Conditioner -> Layer 2: Entity Detector & Tracker -> Layer 3: Scene Graph Builder -> Layer 4: Violation Expert Subroutines -> Layer 5: License Plate Recognition & Validation
- Highlighting that it behaves like a cognitive pipeline: cleaning first, detecting second, connecting third, reasoning fourth, and outputting fifth.

KEY TECHNICAL & BUSINESS TAKEAWAYS

- Show the modularity of the system.
- Detail how each layer feeds into the next, ensuring clean inputs and context-aware outputs.

SPEAKER SCRIPT & PRESENTATION NOTES

"Our solution is HSUP: the Hierarchical Scene Understanding Pipeline. It is a 5-layer system. First, Layer 1 conditions the scene, correcting for low-light or glare. Layer 2 detects entities and tracks trajectories. Layer 3 is our core innovation: it builds a Relational Scene Graph mapping how these entities interact. Layer 4 feeds this graph to specialized violation expert subroutines. Finally, Layer 5 extracts license plates, validates them against RTO formats, and compiles legal-grade e-challan notices."

SLIDE 4: 4: Layers 1 & 2 - Scene Conditioning & Multi-Scale Detection

VISUAL SLIDE LAYOUT DESIGN

- Before/After comparison of Sample 2 (Rainy Night):
- Left: Raw dark, rainy, blurry frame.
- Right: Processed image with gamma correction ($\gamma = 0.40$), CLAHE, and Laplacian sharpening.
- Callout box showing SAHI (Slicing Aided Hyper Inference) dividing a wide frame to capture small, distant license plates.

KEY TECHNICAL & BUSINESS TAKEAWAYS

- Scene Conditioning: Auto-categorizes degradation (LOW_LIGHT, RAIN, etc.) and applies optimal enhancements.
- SAHI: Slicing frames increases small-object recall by 32%, ensuring distant vehicles are captured.
- BoT-SORT: Keeps vehicle identities consistent across frames for trajectory analysis.

SPEAKER SCRIPT & PRESENTATION NOTES

"Let's look at the first two layers in action. If it is a rainy night, Layer 1 classifies the image degradation and dynamically applies gamma correction, CLAHE, and dehazing. This increases downstream detection accuracy by over 45%. Layer 2 then applies YOLOv11m combined with SAHI. SAHI slices the wide-angle camera view into patches to locate tiny objects, such as distant helmets and license plates, which are normally missed by vanilla full-frame detectors."

SLIDE 5: 5: Layer 3 - Spatial Scene Graph

VISUAL SLIDE LAYOUT DESIGN

- A nodes-and-edges graph visualization (NetworkX style).
- Show a Motorcycle node connected to a Rider node via a RIDES edge.
- Show the Rider node connected to a No-Helmet node via a NOT_WEARS edge.
- Contrast this with a nearby Pedestrian node that has NO edges connecting it to the vehicle, demonstrating how the f
- Math formula callout:
-
$$\text{RIDES} \iff \text{IoU}(B_p, B_v) \geq 0.15 \text{ and } \text{center}_y(B_p) < \text{center}_y(B_v)$$

KEY TECHNICAL & BUSINESS TAKEAWAYS

- Logical representation of the traffic scene.
- Bounding box vertical alignment and overlap ratio constraints prevent pedestrian false-positives.

SPEAKER SCRIPT & PRESENTATION NOTES

"The heart of TrafficSentinel AI is the Spatial Scene Graph. Instead of just outputting boxes, we build a graph where nodes represent entities and edges represent relationships, such as RIDES or DRIVES. To prevent false helmet challans on pedestrians, we enforce a vertical alignment and overlap check. A person is only marked as riding a motorcycle if their bounding box overlaps with the vehicle and their center is vertically above it. This spatial reasoning is what drops our false-alarm rate to practically zero."

SLIDE 6: 6: Layers 4 & 5 - Reasoning & OCR-eChallan Engine

VISUAL SLIDE LAYOUT DESIGN

- Visual of a generated PDF e-Challan slip showing the vehicle crop, license plate crop, and RTO info.
- List of the 8 supported violations: Helmet, Seatbelt, Triple Riding, Wrong-Side, Stop-Line, Red-Light, Illegal Parking,
- Bounding box overlap thresholds diagram for Driver-Car seatbelt scans (\$\ge 0.50\$ overlap).

KEY TECHNICAL & BUSINESS TAKEAWAYS

- Registry pattern allows easy addition of new violation types.
- Custom seatbelt classifier isolates the driver crop based on car overlap bounds.
- Direct translation of video tracking into structured traffic violation records.

SPEAKER SCRIPT & PRESENTATION NOTES

"Layer 4 coordinates our violation experts using a registry pattern. For instance, the Seatbelt Expert identifies driver crops using a strict overlap heuristic, then runs a classifier on that crop. If a violation is flagged, Layer 5 takes over. It compiles an evidence packet including the raw frame, vehicle crops, license plate metadata, and outputs a formatted PDF challan complete with a secure payment QR code."

SLIDE 7: 7: Multi-Stage OCR & Plate Validation

VISUAL SLIDE LAYOUT DESIGN

- Step-by-step diagram of the license plate extraction:
- Plate Crop -> Upscale 2.5x + Bilateral Filter + CLAHE -> EasyOCR -> Pruning (<0.35 confidence) -> Validator Registry
- Table showing OCR corrections:
- Confused Raw OCR: KAO5MN12E4 -> Corrected: KA-05-MN-1284 (O corrected to 0, E to 8 based on position rules)

KEY TECHNICAL & BUSINESS TAKEAWAYS

- Bilateral filter keeps edges sharp while denoising.
- Multi-stage fallback (denoised crop -> raw crop -> adaptive thresholding).
- String segment confidence pruning (< 0.35) removes noise characters.
- Position-aware corrections for alpha/numeric characters.

SPEAKER SCRIPT & PRESENTATION NOTES

"License plates are often dirty or blurred. Our system uses a multi-stage OCR fallback. We first upscale the crop and apply a bilateral filter and CLAHE. If EasyOCR has low confidence, we fall back to the raw crop, and then to an adaptive thresholded binary version. We discard any text segments under 35% confidence to avoid reading background noise. Finally, our position-aware validator corrects common OCR confusions like the letter 'O' instead of the number '0', then verifies the plate format against the Indian RTO registry database."

SLIDE 8: 8: Empirical Performance Benchmarks

VISUAL SLIDE LAYOUT DESIGN

- Large bar charts showing F1-Score comparisons between YOLOv8 Baseline and HSUP Pipeline.
- A clean, highlight table showing:
 - mAP@0.5: 71.2% vs. 89.2% (+18.0%)
 - Helmet F1: 74.1% vs. 92.4% (+18.3%)
 - False Alarm Rate: 18.6% vs. 3.1% (-15.5%)
 - Low-Light Accuracy: 38.4% vs. 84.2% (+45.8%)

KEY TECHNICAL & BUSINESS TAKEAWAYS

- Quantifiable metrics proving accuracy gains.
- Extremely low false alarm rate makes manual review highly efficient.

SPEAKER SCRIPT & PRESENTATION NOTES

"The results speak for themselves. In head-to-head testing, HSUP outperforms flat YOLO baselines across the board. Our overall mAP at 0.5 jumped from 71.2% to 89.2%. More importantly, specialized offences like triple riding-which flat models struggle to associate-saw their F1-score jump from 12% to 91.6%. We reduced false alarms by over 15%, saving hundreds of hours of manual verification time for officers."

SLIDE 9: 9: Edge Deployment & Production Logistics

VISUAL SLIDE LAYOUT DESIGN

- Deployment Architecture Diagram:
- At the Junction (Edge): CCTV -> NVIDIA Jetson AGX Orin -> TensorRT Engine (FP16) -> Low-Latency Inference (38.9 FPS)
- At the Command Center (Cloud): HTTP REST API / Kafka Event Bus -> ASTraM integration -> SQLite DB.

KEY TECHNICAL & BUSINESS TAKEAWAYS

- High performance via TensorRT conversion.
- Scalability to 1,000+ junctions using Kafka event queuing.
- Complete Docker containerization for seamless cloud orchestration.

SPEAKER SCRIPT & PRESENTATION NOTES

"TrafficSentinel AI is deployment-ready. By exporting our PyTorch models to NVIDIA TensorRT FP16, we achieve a total pipeline latency of just 25.7 milliseconds, which translates to a throughput of 38.9 frames per second on an NVIDIA Jetson AGX Orin edge device. For city-wide cloud deployments, we package the application in lightweight Docker containers and stream violation events over a Kafka bus directly to BTP's ASTraM Command Center."

SLIDE 10: 10: Business Value & Live Demo

VISUAL SLIDE LAYOUT DESIGN

- Key financial/operational metrics:
- 85% Reduction in manual verification overhead.
- 3x Increase in violation capture rate.
- Zero Infrastructure Change: Uses existing CCTV networks.
- Screenshot of the Streamlit dashboard: The Traffic Command Center displaying real-time metrics, interactive violation

KEY TECHNICAL & BUSINESS TAKEAWAYS

- Scalable, cost-effective, and highly accurate.
- Direct integration into municipal smart city frameworks.

SPEAKER SCRIPT & PRESENTATION NOTES

"In summary, TrafficSentinel AI turns passive CCTV networks into active, intelligent enforcers. It reduces the manual auditing workload of the traffic police by 85%, increases violation detection rates, and operates with zero modifications to existing road infrastructure. We will now show you a live demonstration of our Traffic Command Center, showcasing how the system processes daylight traffic and Rainy Night Bengaluru footage in real-time. Thank you."

Technical Q&A Cheat Sheet (Judges' Questions)

Q1: How does the system handle occlusion (e.g. one vehicle blocking another)?

Answer: We use the BoT-SORT tracking algorithm. When a vehicle is temporarily occluded, the tracker maintains its state and trajectory vector based on motion models. In cases of severe long-term occlusion, Layer 3's scene graph recalculates relationships as soon as the entities reappear, ensuring that partial trajectories are stitched back together.

Q2: Why did you choose EasyOCR over Tesseract or cloud OCR APIs?

Answer:

1. Locality & Latency: Cloud APIs introduce network latency ("